# Sentence Style Detection for Author Switches

## Author Switch Detection

Tshephang Phuti-A-Nkutu Matlala

Academy of Computer Science and Software Engineering
University of Johannesburg
South Africa



1 Problem Statement

2 Representation & Features

3 Models Overview

4 Results and Interpretation

5 Handling Class Imbalance

6 Future Work

7 Conclusion

8 Questions



# Problem statement

**Formally:** for a document $D = \{S_1, \ldots, S_n\}$, predict for every adjacent pair $(S_i, S_{i+1})$:

$$y_i = \begin{cases} 0 & S_i \text{ and } S_{i+1} \text{ share an author,} \\ 1 & \text{an author switch occurs between them.} \end{cases}$$

# Why is it interesting?

- **A difficult problem:** author style is subtle, especially when the topic is controlled
- **Short evidence:** one sentence carries only a small trace of the author
- **Descriptive cues:** short style features (length, punctuation, vocabulary) can still describe an author
- **Interpretable:** we can see which cues drive a decision
- **Useful:** plagiarism detection and analysis of co-authored text



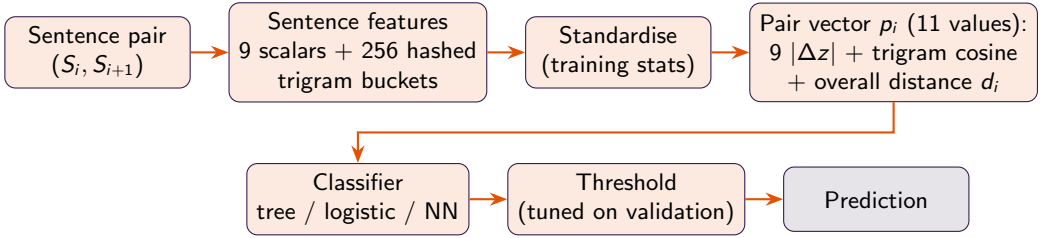
# Task and data

- PAN 2026 data
- Easy: topic changes with author. Medium/hard: topic is controlled
- 2,439,074 training pairs, about 14.3% of them are switches

| | Pairs | Switch rate |
|---|---|---|
| Easy | 549,804 | 30.5% |
| Medium | 1,299,324 | 4.4% |
| Hard | 589,946 | 20.9% |

Switch rates differ sharply by difficulty (easy, medium and hard).

# Pipeline: from raw text to a prediction




- Written in Java with *no external ML library*

# Style features(1/2)

| Idx | Feature | What it captures about the author |
|---|---|---|
| 0 | Word count ($\log(1 + \#\text{words})$) | Preference for short or long sentences |
| 1 | Character count (log scaled) | Sentence length again, but also sensitive to word choice |
| 2 | Frequent word ratio (top 10k English) | Share of common words versus rarer words |
| 3 | Punctuation density | How heavily the author punctuates |
| 4 | Digit ratio | Writing numbers and dates as digits |

Scalar features are z-scored with training statistics.



# Style features (2/2)

| Idx | Feature | What it captures about the author |
|---|---|---|
| 5 | Uppercase ratio | Capitalisation habit: acronyms, emphasis, names |
| 6 | Average word length | Preference for short or long words |
| 7 | Mean TF-IDF | How rare the words are compared with the training text |
| 8 | Emotion score (NRC lexicon) | Emotional wording per word |
| **Not used during training** | | |
| – | Type-token ratio | Word variety versus repetition within a sentence |
| – | Punctuation variety | Range of different punctuation marks used |

# Hashed character trigrams and the pair vector

**Hashing trick**

- Each trigram $\rightarrow$ one of $B = 256$ buckets
- Advantage: fixed length, no corpus required
- Cost: there are collisions, and the buckets mean nothing to us

**Pair feature vector: compressed version (11 values)**

- 9 absolute differences $|z_i^{(k)} - z_{i+1}^{(k)}|$
- Trigram cosine (kept readable)
- Overall distance $d_i$



# Models overview

| Model | Setup |
|---|---|
| Decision tree | Gini, depth 5, min 20 to split, 5 per leaf, class weighted, threshold 0.5 |
| Logistic regression | Batch gradient descent, 10,000 iterations, lr 0.6, class weighted |
| Neural network | 11–128–64–32–16–8–1, ReLU, mini batch SGD (lr 0.001, batch 64), L2 0.01, dropout 0.2, early stopping |



# Results: overall held out test (259,653 pairs)

| Model | Acc. | Prec. | Rec. | F1 | Bal. Acc. |
|---|---|---|---|---|---|
| Decision tree | 0.850 | 0.462 | **0.410** | 0.435 | 0.666 |
| Logistic regression | 0.869 | 0.546 | 0.392 | **0.456** | **0.669** |
| Neural network | **0.871** | **0.562** | 0.368 | 0.445 | 0.661 |

- The three models are nearly tied (the neural network is not better than the linear model)
- The majority baseline already scores **0.859** accuracy, so accuracy alone is barely informative



# Results by difficulty (1/2): easy

F1 / balanced accuracy on the held out test split

| Model | Easy |
|---|---|
| Decision tree | 0.872 / 0.897 |
| Logistic regression | 0.869 / 0.890 |
| Neural network | 0.848 / 0.873 |

- **Easy:** strong, but author changes coincide with topic changes



# Results by difficulty (2/2): medium and hard

F1 / balanced accuracy on the held out test split

| Model | Medium | Hard |
|---|---|---|
| Decision tree | 0.067 / 0.503 | 0.057 / 0.502 |
| Logistic regression | 0.078 / 0.515 | 0.040 / 0.503 |
| Neural network | 0.058 / 0.503 | 0.027 / 0.499 |

- **Medium/hard:** balanced accuracy $\approx 0.50$, i.e. chance

# Interpretation: what the models actually used

**Decision tree**

- Root: character length difference (idx 1)
- Then average word length difference (idx 6)
- Trigram cosine appears at nearly every level
- Only 5 of 11 features used at all

**Logistic regression weights**

- Trigram cosine: $-4.71$ (more similar $\Rightarrow$ fewer switches)
- Character length difference: $+0.99$
- Overall distance: $+0.72$
- Frequent word and punctuation differences $\approx 0$

The nine scalars are mostly surface statistics.



# Handling class imbalance

Only 14.3% of pairs are switches, so “always same author” looks accurate.

## 1. Class weights

Same author **0.583**
Switch **3.505**

## 2. Tuned threshold

Chosen on validation to maximise F1

## 3. Fair metrics

Balanced accuracy, F1 and majority baseline

- No resampling: all 2.4M pairs are kept



# What I am still working on (Future Work)

- **Feature ablation and importance**
- **Restoring n-gram compressed information**
- **Topic robust features:** such as POS tagging usage, readability, vocabulary richness
- **Stronger models:** such as random forest, gradient boosting



# Summary

- Author Switch Detection system: 9 features + hashed trigram profile and distance; 11-value pair feature vector
- The results are baseline F1 (0.435–0.456)



# Thank You

# Questions?

